

Granular-Ball Fuzzy Neighborhood Entropy-Based Outlier Detector

Yongxiang Li, Xinyu Su, Weili Jiang, Gadeng Luosang, Hongmei Chen, and Zhong Yuan

Abstract—Outlier detection is a fundamental task in data mining, aiming to identify samples that significantly deviate from normal data distributions. In real-world applications, such deviations are often embedded in uncertain, noisy, and locally ambiguous data structures, posing substantial challenges to reliable detection. Fuzzy information granulation provides an effective paradigm for modeling such uncertainty by representing data through fuzzy granules and fuzzy neighborhood relationships. However, most existing fuzzy neighborhood-based outlier detection methods are developed within a single-granularity representation framework, which is inherently inconsistent with the hierarchical nature of real-world data, leading to the misinterpretation of local fluctuations as true anomalies. To address these limitations, this paper constructs a hierarchical multi-granularity fuzzy neighborhood representation and proposes a Multi-granularity Granular-ball Fuzzy Neighborhood Entropy (MFNE) outlier detection method. Specifically, by organizing data samples into granular balls at different scales, MFNE reformulates outlier detection from isolated point-wise comparison to structured granule-level uncertainty modeling. Furthermore, MFNE extends fuzzy neighborhood entropy to the multi-granularity granular-ball space and introduces a multi-scale fuzzy neighborhood relative entropy fusion strategy to effectively capture scale-dependent variations in granule uncertainty. Finally, based on the aggregated entropy information, the deviation degree of each granular ball is quantified as its outlier score, which is subsequently propagated back to individual samples to obtain the final detection results. Extensive experiments demonstrate that MFNE outperforms state-of-the-art methods on 16 datasets, validating its effectiveness. The code is publicly available at <https://github.com/LYXRhythm/MFNE.git>.

Index Terms—Granular computing, Fuzzy neighborhood rough sets, Granular-ball computing, Outlier detection

I. INTRODUCTION

Outlier detection, also known as anomaly detection, is a fundamental problem in data mining and machine learning that aims to identify a small number of instances significantly deviating from the majority of the data [1]–[5]. In recent years, unsupervised outlier detection has attracted extensive attention due to its independence from labeled data, and it has been successfully applied in various domains such as financial fraud detection, industrial image, and medical diagnosis. Despite the remarkable progress, effectively characterizing uncertainty and fuzziness within data remains a

critical challenge in this field. Information entropy, as a crucial uncertainty measure in various Granular Computing (GrC) models, provides a quantitative means to assess the degree of disorder and the discriminative power of data distributions. Under normal conditions, data typically follow relatively stable and consistent patterns, whereas the presence of outliers disrupts this stability and leads to a significant increase in overall uncertainty. As a result, information entropy can effectively capture the distributional perturbations introduced by outliers, making it a powerful tool for revealing hidden abnormal patterns. Owing to this property, entropy has been widely employed in outlier detection tasks and has demonstrated strong effectiveness across various application domains [6].

Existing outlier detection methods can be broadly categorized into several classes. Statistical-based methods [7] identify outliers by modeling the underlying probability distribution of data and detecting samples that deviate from the assumed distribution. These methods are theoretically sound and easy to implement, but their performance is often limited when dealing with data with an unknown distribution. Distance-based methods [8] determine outliers by measuring pairwise distances between samples, offering strong intuition but suffering from the curse of dimensionality and high computational cost in large datasets. Density-based methods [9], [10] detect outliers by comparing local density variations with neighbors. However, their effectiveness heavily depends on parameter tuning and neighborhood size. Recently, deep learning-based methods [4], [11] have attracted significant attention due to their powerful representation learning ability, enabling effective detection in complex data scenarios. Nevertheless, such methods typically require large training datasets and lack interpretability.

In contrast, rough sets-based outlier detection methods provide an effective way to handle uncertainty and fuzziness in data without requiring prior knowledge of probability distributions or additional assumptions. Rough set theory characterizes boundary regions through lower and upper approximations, offering a natural framework for identifying borderline objects and outliers. However, classical rough sets cannot effectively handle numerical data, leading to the development of various extended rough set models to improve their applicability to numerical and mixed data types, such as Fuzzy Rough Sets (FRS) [12], Neighborhood Rough Sets (NRS) [13], Kernelized Fuzzy Rough Sets (KFRS) [1], [14], and Fuzzy Neighborhood Rough Sets (FNRS) [15]. These models introduce multiple uncertainty measures to uncover abnormal information in data, including fuzzy approximation accuracy [12], fuzzy information entropy [16], neighborhood entropy [13], and fuzzy rough

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density [17]. As a result, rough set-based methods have gained increasing attention in recent years and demonstrated unique advantages in addressing uncertainty-related challenges.

To address the limitation of NRS in handling fuzziness, Wang et al. [15] first combined the strengths of FRS and NRS to develop FNRS, providing a novel model capable of effectively handling uncertainty information, which was successfully applied to attribute reduction with promising results. Subsequently, Yuan et al. [18] extended FNRS to a decision-free fuzzy neighborhood information system, using granule intensity to reflect the outlier degrees of samples. Deng et al. [19] enhanced outlier detection performance by leveraging multiple information extraction based on FNRS. Yuan et al. [9] combined FNRS with the density-based Local Outlier Factor (LOF) method, calculating sample outlier scores through fuzzy neighborhood deviation degrees. However, these methods rely on a fixed paradigm that uses single-granularity samples as the basic processing unit, requiring the computation of fuzzy relations between single-granularity samples. This results in low efficiency, sensitivity to noise, and an inability to effectively utilize multi-granularity information in data, leading to suboptimal outlier detection. Furthermore, the application of FNRS in outlier detection is currently limited, and more uncertainty measures suitable for outlier detection warrant further exploration and research.

Granular-Ball Computing (GBC) is a novel multi-granularity representation and computation model in GrC. In recent years, GBC has been widely applied in outlier detection, demonstrating promising results in both detection efficiency and performance [5], [20]. For instance, Su et al. [20] combined FRS with GBC to improve the computational efficiency and multi-granularity information utilization of traditional FRS, introducing granular-ball fuzzy approximation accuracy for outlier detection. Cheng et al. [21] integrated GBC into a traditional mean shift-based method, enhancing efficiency and robustness through multi-granularity granular-balls. Li et al. [22] combined GBC with fuzzy set theory, detecting outliers by fusing granular-ball fuzzy information. Gao et al. [5] integrated FRS with multi-scale granular-balls, innovatively transforming unsupervised outlier detection into a semi-supervised classification problem. GBC replaces the traditional fixed paradigm based on single-granularity samples with a multi-granularity paradigm based on granular-balls, providing an effective tool to address the above issues.

Information fusion aims to integrate complementary information from different perspectives or levels to obtain more comprehensive data representations, thereby alleviating the limitations caused by incomplete or biased characterizations in a single feature space. Information fusion has been widely applied in outlier detection, with primary fusion strategies including multi-source information fusion [23], [24], multi-scale information fusion [5], [25], and multi-granularity information fusion [10]. Among these, multi-granularity information fusion aligns closely with the concept of GrC. Coarse-grained representations reduce the impact of noise and redundant information, enhancing model generalization, while fine-grained representations preserve local structures and individual differences, aiding in the identification of potential

outlier patterns.

Based on the above discussion, we propose a Multi-granularity granular-ball Fuzzy Neighborhood Entropy-based outlier detector (MFNE). The framework of MFNE is presented in Fig. 1. In MFNE, we integrate GBC with FNRS to develop a multi-granularity fuzzy neighborhood rough sets model. Initially, we generate multi-granularity granular-balls on raw data to achieve a multi-granularity representation. Subsequently, the data is granulated into multiple multi-granularity fuzzy neighborhood information granules. To capture abnormal information in the data, we introduce several novel uncertainty measures, including multi-granularity fuzzy neighborhood entropy, fuzzy neighborhood relative entropy, and fuzzy neighborhood relative cardinality. To better fuse uncertainty information in data for enhancing detection performance, we introduce a novel multi-granularity information fusion strategy, termed the multi-granularity fuzzy neighborhood relative entropy fusion strategy. Using this fusion approach, we leverage the multi-granularity and uncertainty information in data by fusing the multi-granularity neighborhood relative entropy. Finally, we calculate the outlier scores of granular-balls by aggregating the deviation degrees of multiple multi-granularity fuzzy neighborhood information granules and map these scores back to individual samples within the granular-balls to obtain their outlier scores. Outliers are identified by applying a threshold to each sample's outlier score.

In summary, the contributions of this study are as follows.

- 1) We pioneeringly propose a unified framework integrating GBC with FNRS, introducing a multi-granularity fuzzy neighborhood rough sets model.
- 2) We introduce a novel multi-granularity information fusion strategy that leverages multi-granularity fuzzy neighborhood relative entropy to utilize uncertainty information in data, enhancing outlier detection performance.
- 3) We propose a Multi-granularity granular-ball Fuzzy Neighborhood Entropy-based outlier detector (MFNE), which combines GBC and FNRS to improve the efficiency and effectiveness of outlier detection.
- 4) Experimental results demonstrate that MFNE outperforms existing methods on multiple real-world datasets, validating its effectiveness.

II. RELATED WORK

A. Rough sets-based outlier detection

Traditional rough set-based methods are limited in application due to their ability to handle only nominal attribute data. To effectively manage uncertainty information in heterogeneous data, various extended rough set models and new uncertainty measures have been proposed to enhance the adaptability and effectiveness of rough set theory in outlier detection. For instance, Yuan et al. [12] applied FRS to unsupervised outlier detection, using the fuzzy approximation accuracy of fused fuzzy rough granules as an uncertainty measure to reflect the outlier degrees of samples. Yuan et al. [17], [26] extended FRS to outlier detection in mixed-attribute data, calculating outlier scores based on fuzzy rough density. Wu et al. [1]

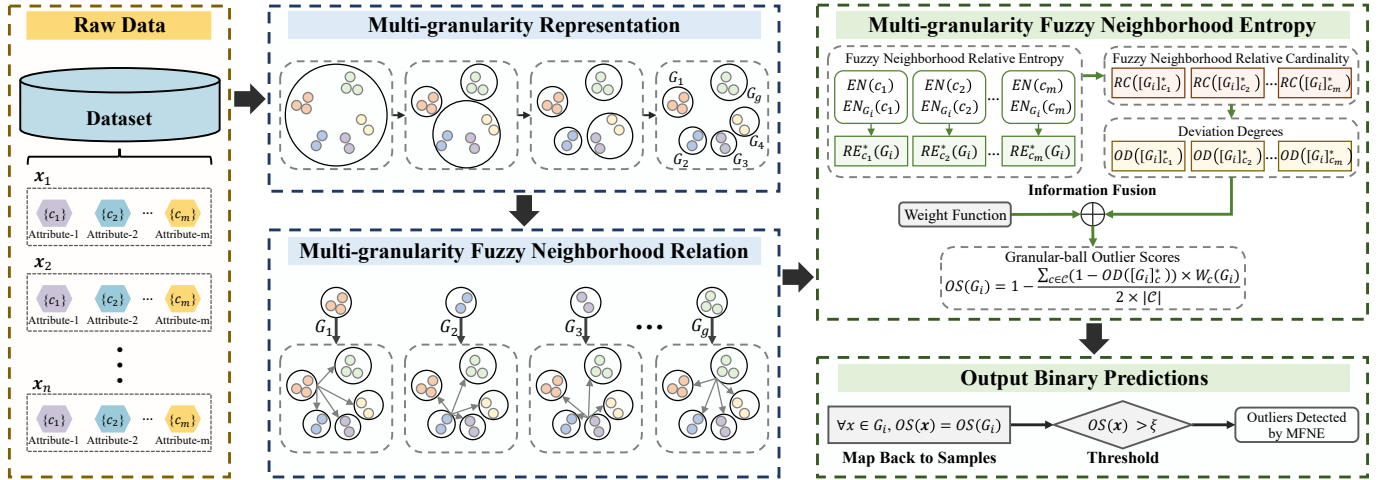


Fig. 1: The framework of MFNE.

utilized KFRS for unsupervised outlier detection, detecting outliers by fusing kernelized fuzzy approximation accuracy of fuzzy information granules. Su et al. [20] pioneered the integration of GBC and FRS for outlier detection, proposing the use of granular-ball fuzzy approximation accuracy to identify outliers. Chen et al. [25] proposed a multi-scale information fusion model based on FRS, detecting outliers through the fuzzy approximation accuracy of multi-scale fuzzy information granules. Cai et al. [27] proposed a prototype-based FRS method, performing outlier detection through prototype learning based on separable attribute selection. Liao et al. [28] introduced optimistic and pessimistic fuzzy granule densities based on FRS to assess density changes in local neighborhoods, while also incorporating a mean-shift-based distance metric to detect global and group outliers. Yuan et al. [18] applied FNRS to unsupervised outlier detection, using the ratio of fuzzy neighborhood upper and lower approximations to reflect the outlier degrees of samples.

FNRS, by integrating FRS and NRS, offers a novel framework for handling uncertainty in outlier detection. While FNRS has been explored in the context of outlier detection, current FNRS-based approaches remain constrained by several inherent shortcomings. These methods adopt a fixed paradigm that uses single-granularity samples as the basic processing unit, requiring at least $O(n^2)$ time complexity to calculate fuzzy relations (where n is the number of samples), resulting in low time efficiency. Additionally, this single-granularity processing paradigm leads to sensitivity to noise and an inability to effectively utilize multi-granularity information. Therefore, further exploration of FNRS in the field of outlier detection is necessary.

B. Granular-ball computing

GBC, as a recently proposed key GrC model, has been successfully applied in various domains of machine learning and deep learning. Granular-balls, as the core data structure and basic processing unit in GBC, are defined by a set containing multiple samples, characterized by two key attributes, center and radius. Given a granular-ball $G =$

$\{\mathbf{x}_1, \mathbf{x}_2, \dots, \mathbf{x}_l\}$, its center is calculated as $\mathbf{c} = \frac{1}{l} \sum_{i=1}^l \mathbf{x}_i = (\frac{1}{l} \sum_{i=1}^l x_i^1, \dots, \frac{1}{l} \sum_{i=1}^l x_i^m)$, and its radius is calculated as $r = \frac{1}{l} \sum_{i=1}^l \|\mathbf{x}_i - \mathbf{c}\| = \frac{1}{l} \sum_{i=1}^l \sqrt{\sum_{j=1}^m (x_i^j - c^j)^2}$, where l denotes the number of samples in G , m denotes the number of attributes, x_i^j denotes the j -th attribute of the sample $\mathbf{x}_i$, c^j denotes the j -th attribute of the center $\mathbf{c}$, and $\|\cdot\|$ denotes the 2-norm. Due to the coarse-grained nature of granular-balls, even if a granular-ball contains noise, its adverse impact on samples is mitigated by the smoothing effect of other samples within the same granular-ball, thereby enhancing the robustness of subsequent models during processing [29].

The quality of generated granular-balls directly impacts the effectiveness of subsequent tasks, making an efficient granular-ball generation method crucial. For example, Xia et al. [30] improved the traditional k -means-based generation method through division, accelerating the generation process while ensuring granular-ball quality. Zhang et al. [31] proposed a fast adaptive granular-ball generation method that reduces generation time and adaptively adjusts the granular-ball radius. Xie et al. [32] addressed the instability of division-based and k -means-based generation methods by proposing a method that calculates the distance from data-driven centers to unpartitioned samples during granular-ball division, eliminating the need for random center selection and distance calculations to all samples, thereby improving generation efficiency and quality. Yuan et al. [33] introduced two innovative quantum granular-ball generation methods, leveraging the inherent properties of quantum computing. Liu et al. [34] proposed a local density-based generation method that selects granular-ball centers based on local density, ensuring that the generated granular-balls better align with the original data distribution.

These efficient, robust, and adaptive granular-ball generation methods provide a foundation for the performance of various subsequent tasks, enabling GBC to be applied across diverse applications. For example, Chen et al. [35] proposed a granular-ball-based selector for attribute reduction, facilitating data dimensionality reduction. Xia et al. [36] introduced a granular-ball sampling method that employs adaptively gener-

ated granular-balls to cover the data space, with points within the granular-balls forming the sampled data. This method reduces data scale while improving data quality. Cheng et al. [37] introduced a GBC-based density peak clustering method, improving the efficiency of the original method. Xia et al. [38] applied GBC to rough set theory, proposing granular-ball rough sets that can handle continuous data and use equivalence classes for knowledge representation. Wang et al. [39] utilized granular-balls to fit samples and construct a state transition matrix for random walks, generating a steady-state distribution through iterative computation for outlier detection. Su et al. [40] improved the original granular-ball generation method to handle mixed-attribute data, combining granular-ball fuzzy sets for outlier detection.

Despite the achievements of GBC in outlier detection, exploration of new uncertainty measures and fusion strategies in data remains limited, and its application in outlier detection warrants further investigation and research. In this study, we focus on mining and fusing uncertainty information in data to enhance outlier detection performance. To address the shortcomings of existing methods, we propose, for the first time, a unified framework integrating GBC and FNRS, introducing a multi-granularity fuzzy neighborhood rough sets model. To more effectively utilize uncertainty information in data, we develop a novel multi-granularity information fusion strategy that distinguishes inliers and outliers by fusing multi-granularity fuzzy neighborhood relative entropy. Finally, we propose MFNE, which quantifies the outlier degrees of samples by leveraging the deviation degrees of multiple multi-granularity fuzzy neighborhood information granules.

III. PRELIMINARIES

In this section, we review the foundational concepts of FNRS and several uncertainty measures. Let $U = [x_1; x_2; \dots; x_n] \in \mathbb{R}^{n \times m}$ denote the data matrix with n samples and m attributes; each sample $x_i \in U$ is denoted as $x_i = (x_i^1, x_i^2, \dots, x_i^m) \in \mathbb{R}^m$, x_i^j denotes the j -th attribute of x_i . Let $\mathcal{C}$ denote the conditional attribute set, for any $\mathcal{B} \subseteq \mathcal{C}$, a fuzzy relation $\mathcal{R}_{\mathcal{B}}$ induced by $\mathcal{B}$ on U can be defined as a mapping $\mathcal{R}_{\mathcal{B}} : U \times U \rightarrow [0, 1]$. $\mathcal{R}_{\mathcal{B}}$ can also be denoted as a fuzzy relation matrix $\mathbf{R}_{\mathcal{B}} = [r_{ij}]_{n \times n}$, where $r_{ij} = \mathcal{R}_{\mathcal{B}}(x_i, x_j)$ denotes the fuzzy relation between x_i and x_j . For any $x_i, x_j \in U$, if $\mathcal{R}_{\mathcal{B}}$ satisfies (1) reflexivity i.e. $\mathcal{R}_{\mathcal{B}}(x_i, x_i) = 1$ and (2) symmetry i.e. $\mathcal{R}_{\mathcal{B}}(x_i, x_j) = \mathcal{R}_{\mathcal{B}}(x_j, x_i)$, then $\mathcal{R}_{\mathcal{B}}$ is called a fuzzy similarity relation.

Definition 1. Given a neighborhood radius ε , for any $\mathcal{B} \subseteq \mathcal{C}$ and any $x_i, x_j \in U$, the fuzzy neighborhood information granule of x_i w.r.t. $\mathcal{B}$ is defined as

$$[x_i]_{\mathcal{B}}^{\varepsilon}(x_j) = \begin{cases} \mathcal{R}_{\mathcal{B}}(x_i, x_j), & \mathcal{R}_{\mathcal{B}}(x_i, x_j) \geq \varepsilon; \\ 0, & \mathcal{R}_{\mathcal{B}}(x_i, x_j) < \varepsilon. \end{cases} \quad (1)$$

For any $\mathcal{B} \subseteq \mathcal{C}$, the fuzzy neighborhood information granules induced by $\mathcal{B}$ for each sample form a fuzzy neighborhood granular structure w.r.t. $\mathcal{B}$, denoted as $S(\mathcal{R}_{\mathcal{B}}) = \{[x_1]_{\mathcal{B}}^{\varepsilon}, [x_2]_{\mathcal{B}}^{\varepsilon}, \dots, [x_n]_{\mathcal{B}}^{\varepsilon}\}$. The cardinality of $[x_i]_{\mathcal{B}}^{\varepsilon}$ is calculated as $|[x_i]_{\mathcal{B}}^{\varepsilon}| = \sum_{x_j \in U} [x_i]_{\mathcal{B}}^{\varepsilon}(x_j)$. For any $\mathcal{B}_1, \mathcal{B}_2 \subseteq \mathcal{C}$ and any $x_i \in U$, we can obtain the following properties [18].

- 1) If $\mathcal{B}_1 \subseteq \mathcal{B}_2$, then $\mathcal{R}_{\mathcal{B}_2} \subseteq \mathcal{R}_{\mathcal{B}_1}$;
- 2) If $\mathcal{B}_1 \subseteq \mathcal{B}_2$, then $[x_i]_{\mathcal{B}_2}^{\varepsilon} \subseteq [x_i]_{\mathcal{B}_1}^{\varepsilon}$;
- 3) If $\varepsilon_1 \leq \varepsilon_2$, then $[x_i]_{\mathcal{B}_1}^{\varepsilon_2} \subseteq [x_i]_{\mathcal{B}_1}^{\varepsilon_1}$.

Based on the fuzzy neighborhood information granules defined above, we introduce several common uncertainty measures, including fuzzy neighborhood upper, lower approximations and the fuzzy neighborhood lower approximation ratio.

Definition 2. Given a neighborhood radius ε , for any $\mathcal{B}_1, \mathcal{B}_2 \subseteq \mathcal{C}$ and any $x_i, x_j \in U$, the fuzzy neighborhood lower and upper approximations of $[x_i]_{\mathcal{B}_1}^{\varepsilon}$ w.r.t. $\mathcal{B}_2$ are respectively defined as

$$\underline{\mathcal{N}}_{\mathcal{B}_2}^{\varepsilon}([x_i]_{\mathcal{B}_1}^{\varepsilon}) = \{x_j \in U \mid [x_j]_{\mathcal{B}_2}^{\varepsilon} \subseteq [x_i]_{\mathcal{B}_1}^{\varepsilon}\}, \quad (2)$$

$$\overline{\mathcal{N}}_{\mathcal{B}_2}^{\varepsilon}([x_i]_{\mathcal{B}_1}^{\varepsilon}) = \{x_j \in U \mid [x_j]_{\mathcal{B}_2}^{\varepsilon} \cap [x_i]_{\mathcal{B}_1}^{\varepsilon} \neq \emptyset\}. \quad (3)$$

IV. FUZZY NEIGHBORHOOD ENTROPY GUIDED MULTI-GRANULARITY FUSION FOR OUTLIER DETECTION

A. Detection method

1) *Multi-granularity representation:* In this study, we employ GBC to achieve an adaptive multi-granularity representation of the input raw dataset. The granular-ball generation process essentially covers the original dataset with multiple multi-granularity granular-balls, simulating the human brain's cognitive process from coarse to fine granularity. We adopt an adaptive, efficient, and widely used granular-ball generation method for outlier detection [41]. This method initially treats the entire dataset as a single coarse-grained granular-ball, then selects the two farthest samples as ball centers to divide it into two sub-balls. Subsequently, the method evaluates the quality of each granular-ball, where the quality measure of a granular-ball G is calculated as

$$Q(G) = \frac{|G|}{\sum_{x_t \in G} \|x_t - c\|}, \quad (4)$$

where $|G|$ denotes the number of samples in G . To ensure the effective division of granular-balls, this method also calculates the weighted quality of the sub-balls after division, which is calculated as

$$Q^*(G) = \frac{|G_1|}{|G|}Q(G_1) + \frac{|G_2|}{|G|}Q(G_2), \quad (5)$$

where G_1, G_2 denotes the two granular-balls generated by dividing G . $Q^*(G)$ reflects the weighted quality of the sub-balls formed after the division of G . If $Q^*(G) > Q(G)$, then G is divided into G_1 and G_2 .

Additionally, to address the issue of excessively large granular-ball radii caused by boundary points or noise in the data, this method further adaptively divides relatively coarse granular-balls. Given a granular-ball G_z , if $r_z > 2 \times \max(\text{mean}(r), \text{median}(r))$, G_z will be further divided, where $\text{mean}(r)$ and $\text{median}(r)$ denote the mean and median of the radii of all granular-balls, respectively.

It should be noted that selecting the two furthest samples as initial splitting seeds is a heuristic strategy and may be affected by extreme or isolated noise points. If the selected seeds are noise points, the candidate split may be biased, possibly producing a noise-dominated small sub-ball or an

inappropriate partition of the normal region. In the adopted generation process, these samples are only temporary seeds, and the center and radius of each sub-ball are recalculated after sample assignment. Moreover, a split is accepted only when $Q^*(G) > Q(G)$; otherwise, the current granular-ball is temporarily retained as a coarse-grained representation. If this retained granular-ball still has an excessively large radius, the subsequent radius-based division step can further split it. Nevertheless, more robust initialization strategies can be considered in future work.

Algorithm 1: Granular-ball generation

Input: Dataset U .
Output: A set of generated granular-balls $\mathcal{G}$.

```

1 Initialize the granular-balls  $\mathcal{G} \leftarrow \{U\}$ ;
2 for  $G_z$  in  $\mathcal{G}$  do
3   calculate  $Q(G_z)$  and  $Q^*(G_z)$  by Eqs. (4) and (5);
4   if  $Q^*(G_z) > Q(G_z)$  then
5     Divide  $G_z$  into  $G_{z_1}$  and  $G_{z_2}$ ;
6     Remove  $G_z$  from  $\mathcal{G}$ ;
7     Save granular-balls  $\mathcal{G} \leftarrow \mathcal{G} \cup \{G_{z_1}, G_{z_2}\}$ ;
8   end
9 end
10 Calculate the mean and median of the radii of all granular-balls in  $\mathcal{G}$ ;
11 for  $G_z$  in  $\mathcal{G}$  do
12   if  $r_z > 2 \times \max(\text{mean}(r), \text{median}(r))$  then
13     Divide  $G_z$  into  $G_{z_1}$  and  $G_{z_2}$ ;
14     Remove  $G_z$  from  $\mathcal{G}$ ;
15     Save granular-balls  $\mathcal{G} \leftarrow \mathcal{G} \cup \{G_{z_1}, G_{z_2}\}$ ;
16   end
17 end
18 return Generated granular-balls  $\mathcal{G}$ .
```

In summary, the complete granular-ball generation algorithm is presented as Algorithm 1, with a time complexity of $O(n \log n)$ [41], where n denotes the number of samples. With the aid of this algorithm, the original sample space is re-represented as a multi-granularity granular-ball space, providing a new basic processing unit for the subsequently proposed MFNE.

2) *Multi-granularity fuzzy neighborhood rough sets:* According to the previous definitions, the traditional FNRS model requires the calculation of a fuzzy relation matrix between all pairs of samples. This results in an $O(n^2)$ time complexity, which is computationally prohibitive for large-scale datasets. Furthermore, this fine-grained processing paradigm is often overly sensitive to noise and fails to exploit multi-granularity information inherent in the data. To mitigate these issues, we integrate GBC with the original FNRS to propose an enhanced multi-granularity fuzzy neighborhood rough sets model. Specifically, we first define the fuzzy relations between individual samples and then extend these definitions to the GBC framework. This allows for the construction of a multi-granularity fuzzy relation between granular-balls, effectively transforming the computational focus from individual samples to multi-granularity granular-balls.

Definition 3. For any $b \in \mathcal{C}$ and any $x_i, x_j \in U$, the fuzzy relation between x_i and x_j is defined as

$$\mathcal{R}_b(x_i, x_j) = 1 - \Delta_b(x_i, x_j), \quad (6)$$

where Δ_b denotes the Mahalanobis distance under attribute b . For any $\mathcal{B} \subseteq \mathcal{C}$, the fuzzy relation $\mathcal{R}_{\mathcal{B}}$ is calculated by $\mathcal{R}_{\mathcal{B}} = \cap_{b \in \mathcal{B}} \mathcal{R}_b$.

It should be noted that $\mathcal{R}_{\mathcal{B}} = \cap_{b \in \mathcal{B}} \mathcal{R}_b$ provides a general definition of the fuzzy relation induced by an attribute subset $\mathcal{B}$. Directly intersecting a large number of attribute-wise fuzzy relations may reduce the overall similarity in high-dimensional datasets. To avoid this potential degeneration, the proposed MFNE does not directly construct the final outlier score from a single fuzzy relation induced by the full attribute set $\mathcal{C}$. Instead, in the detection stage, MFNE mainly uses attribute-induced fuzzy neighborhood information granules under each individual attribute $c \in \mathcal{C}$ and then fuses their entropy-based deviation degrees.

Furthermore, we define the multi-granularity fuzzy relation between granular-balls through their centers.

Definition 4. Given the granular-balls $\mathcal{G} = \{G_1, G_2, \dots, G_g\}$ generated on U , for any $b \in \mathcal{C}$ and any $G_i, G_j \in \mathcal{G}$, the multi-granularity fuzzy relation between G_i and G_j is defined as

$$\mathcal{R}_b^*(G_i, G_j) = \mathcal{R}_b(c_i, c_j), \quad (7)$$

where c_i and c_j denotes the centers of G_i and G_j , respectively.

Obviously, the multi-granularity fuzzy relation $\mathcal{R}_b^*$ w.r.t. b on $\mathcal{G}$ is a fuzzy set on $\mathcal{G} \times \mathcal{G}$ and $\mathcal{R}_b^*$ can be denoted as a multi-granularity fuzzy relation matrix, i.e., $\mathbf{R}_b^* = [r_{ij}^*]_{|\mathcal{G}| \times |\mathcal{G}|}$, where $r_{ij}^* = \mathcal{R}_b^*(G_i, G_j)$. Each row vector in $\mathbf{R}_b^*$ denotes a multi-granularity fuzzy set.

Based on the multi-granularity fuzzy relation defined above, we can define the multi-granularity fuzzy neighborhood information granules of granular-balls w.r.t. different attribute subsets as follows.

Definition 5. Given the granular-balls $\mathcal{G} = \{G_1, G_2, \dots, G_g\}$ generated on U , for any $\mathcal{B} \subseteq \mathcal{C}$ and any $G_i, G_j \in \mathcal{G}$, the multi-granularity fuzzy neighborhood information granule of G_i w.r.t. $\mathcal{B}$ is defined as

$$[G_i]_{\mathcal{B}}^*(G_j) = \begin{cases} \mathcal{R}_{\mathcal{B}}^*(G_i, G_j), & \mathcal{R}_{\mathcal{B}}^*(G_i, G_j) \geq \varepsilon_{\mathcal{B}}; \\ 0, & \mathcal{R}_{\mathcal{B}}^*(G_i, G_j) < \varepsilon_{\mathcal{B}}, \end{cases} \quad (8)$$

where $\varepsilon_{\mathcal{B}}$ denotes the neighborhood radius w.r.t. $\mathcal{B}$.

In the above definition, we first use the neighborhood radius $\varepsilon_{\mathcal{B}}$ to determine the multi-granularity neighborhood of a granular-ball G_i . Unlike traditional NRS, where the neighborhood relations between G_i and other neighborhood granular-balls are uniformly treated as 1, in the multi-granularity neighborhood, these relations are characterized by multi-granularity fuzzy relations, forming multi-granularity fuzzy neighborhood information granules. This approach enables a more reasonable definition of multi-granularity neighborhood relations between granular-balls and provides a more accurate characterization of their interactions. The quality of the neighborhood radius $\varepsilon_{\mathcal{B}}$ significantly affects the quality of the constructed neighborhood. Therefore, we adjust the neighborhood radius for each

attribute subset. For any $\mathcal{B} \subseteq \mathcal{C}$, the neighborhood radius w.r.t. $\mathcal{B}$ is calculated as $\varepsilon_{\mathcal{B}} = \text{std}(\mathcal{B})/\sigma$, where $\text{std}(\mathcal{B})$ denotes the standard deviation of $\mathcal{B}$, and σ is an adjustable hyperparameter used to control the size of the neighborhood radius.

Similarly, for any $\mathcal{B} \subseteq \mathcal{C}$, we can define the multi-granularity fuzzy neighborhood information granules for each granular-ball under the attribute subset $\mathcal{B}$. These information granules collectively form the multi-granularity fuzzy neighborhood granular structure w.r.t. $\mathcal{B}$, which can be denoted as

$$M(\mathcal{R}_{\mathcal{B}}^*) = \{[G_1]_{\mathcal{B}}^*, [G_2]_{\mathcal{B}}^*, \dots, [G_g]_{\mathcal{B}}^*\}, \quad (9)$$

where $[G_i]_{\mathcal{B}}^*$ denotes the multi-granularity fuzzy neighborhood information granule of G_i w.r.t. $\mathcal{B}$. The cardinality of $[G_i]_{\mathcal{B}}^*$ is calculated as $|[G_i]_{\mathcal{B}}^*| = \sum_{G_j \in \mathcal{G}} [G_i]_{\mathcal{B}}^*(G_j)$ and $1 \leq |[G_i]_{\mathcal{B}}^*| \leq g$, where g denotes the number of granular-balls generated on U .

Next, based on the defined multi-granularity fuzzy neighborhood information granules, we introduce several uncertainty measures. Information entropy reflects the disorder or randomness of data distribution, which can be regarded as a manifestation of outliers. Therefore, we introduce multi-granularity fuzzy neighborhood entropy to measure the inherent uncertainty of multi-granularity fuzzy neighborhood information granules.

Definition 6. Given the granular-balls $\mathcal{G} = \{G_1, G_2, \dots, G_g\}$ generated on U and for any $\mathcal{B} \subseteq \mathcal{C}$, we can obtain the multi-granularity fuzzy neighborhood granular structure $M(\mathcal{R}_{\mathcal{B}}^*) = \{[G_1]_{\mathcal{B}}^*, [G_2]_{\mathcal{B}}^*, \dots, [G_g]_{\mathcal{B}}^*\}$. The multi-granularity fuzzy neighborhood entropy w.r.t. $\mathcal{R}_{\mathcal{B}}^*$ on $\mathcal{G}$ is defined as

$$EN(\mathcal{R}_{\mathcal{B}}^*) = EN(\mathcal{B}) = -\frac{1}{|\mathcal{G}|} \sum_{[G_i]_{\mathcal{B}}^* \in M(\mathcal{R}_{\mathcal{B}}^*)} \log_2 \frac{|[G_i]_{\mathcal{B}}^*|}{|\mathcal{G}|}. \quad (10)$$

Similarly, we extend the fuzzy neighborhood lower and upper approximations and the fuzzy neighborhood lower approximation ratio, originally defined in the FNRS model, to our proposed multi-granularity fuzzy neighborhood rough sets model.

Definition 7. Given the granular-balls $\mathcal{G} = \{G_1, G_2, \dots, G_g\}$ generated on U , for any $\mathcal{B}_1, \mathcal{B}_2 \subseteq \mathcal{C}$ and any $G_i, G_j \in \mathcal{G}$, the multi-granularity fuzzy neighborhood lower and upper approximations of $[G_i]_{\mathcal{B}_1}^*$ w.r.t. $\mathcal{B}_2$ are respectively defined as

$$\underline{\mathcal{M}}_{\mathcal{B}_2}^*([G_i]_{\mathcal{B}_1}^*) = \{G_j \in \mathcal{G} \mid [G_j]_{\mathcal{B}_2}^* \subseteq [G_i]_{\mathcal{B}_1}^*\}, \quad (11)$$

$$\overline{\mathcal{M}}_{\mathcal{B}_2}^*([G_i]_{\mathcal{B}_1}^*) = \{G_j \in \mathcal{G} \mid [G_j]_{\mathcal{B}_2}^* \cap [G_i]_{\mathcal{B}_1}^* \neq \emptyset\}. \quad (12)$$

Thus far, by integrating GBC with FNRS and adopting multi-granularity granular-balls as the basic processing unit to replace the outdated single-granularity sample-based unit, we have constructed a multi-granularity fuzzy neighborhood rough sets model. Additionally, we have introduced several new uncertainty measures, laying the theoretical foundation for subsequent outlier detection.

3) *Outlier detection:* Based on our proposed multi-granularity fuzzy neighborhood rough sets model, we quantify the outlier degrees of granular-balls by fusing multi-granularity fuzzy neighborhood relative entropy. Subsequently, we calculate the outlier scores of each granular-ball by aggregating the deviation degrees of multiple multi-granularity fuzzy neighborhood information granules to which the granular-ball belongs, and map these scores back to the samples within the granular-ball to obtain their outlier scores.

The multi-granularity fuzzy neighborhood entropy defined above provides a comprehensive uncertainty description for different attribute subsets. Building on this concept, we construct a new uncertainty measure to reflect the uncertainty of each granular-ball. Specifically, given the granular-balls $\mathcal{G} = \{G_1, G_2, \dots, G_g\}$ generated on U , we remove a granular-ball G_i from $\mathcal{G}$ and construct a new multi-granularity fuzzy neighborhood granular structure $M'(\mathcal{R}_{\mathcal{B}}^*)$ based on the remaining granular-balls. The multi-granularity fuzzy neighborhood entropy w.r.t. $\mathcal{R}_{\mathcal{B}}^*$ on $\mathcal{G} - \{G_i\}$ is then defined as follows.

Definition 8. After removing G_i from $\mathcal{G}$, the multi-granularity fuzzy neighborhood entropy w.r.t. $\mathcal{R}_{\mathcal{B}}^*$ on $\mathcal{G} - \{G_i\}$ is defined as

$$\begin{aligned} EN_{G_i}(\mathcal{R}_{\mathcal{B}}^*) &= EN_{G_i}(\mathcal{B}) \\ &= -\frac{1}{|\mathcal{G} - \{G_i\}|} \sum_{[G_j]_{\mathcal{B}}^* \in M'(\mathcal{R}_{\mathcal{B}}^*)} \log_2 \frac{|[G_j]_{\mathcal{B}}^*|}{|\mathcal{G} - \{G_i\}|}. \end{aligned} \quad (13)$$

Furthermore, we can calculate the multi-granularity fuzzy neighborhood relative entropy of the granular-ball G_i w.r.t. $\mathcal{R}_{\mathcal{B}}^*$ as follows.

Definition 9. Given the granular-balls $\mathcal{G} = \{G_1, G_2, \dots, G_g\}$ generated on U and for any $G_i \in \mathcal{G}$, the multi-granularity fuzzy neighborhood relative entropy of G_i w.r.t. $\mathcal{R}_{\mathcal{B}}^*$ is defined as

$$RE_{\mathcal{B}}^*(G_i) = \begin{cases} 1 - \frac{EN_{G_i}(\mathcal{B})}{EN(\mathcal{B})}, & EN_{G_i}(\mathcal{B}) < EN(\mathcal{B}); \\ 0, & EN_{G_i}(\mathcal{B}) \geq EN(\mathcal{B}), \end{cases} \quad (14)$$

In the above definition, we capture the uncertainty of the granular-ball G_i by measuring the change in entropy before and after its removal, thereby providing a feature to reflect the outlier degree of G_i . When G_i is removed, if the entropy of $\mathcal{R}_{\mathcal{B}}^*$ decreases, it indicates that G_i contributes higher uncertainty to $\mathcal{R}_{\mathcal{B}}^*$ and is more likely to be an outlier. Conversely, if the entropy change is minimal or increases, it suggests that G_i has relatively low uncertainty and is more likely to be an inlier. Therefore, we can quantify the uncertainty of G_i using the multi-granularity fuzzy neighborhood relative entropy, where a larger value indicates higher uncertainty and a greater likelihood of G_i being an outlier.

Considering that different multi-granularity fuzzy neighborhood information granules may exhibit varying outlier degrees, we further introduce the multi-granularity fuzzy neighborhood relative cardinality to reflect whether each information granule represents a major class or a minority class.

Definition 10. Given the granular-balls $\mathcal{G} = \{G_1, G_2, \dots, G_g\}$ generated on $\mathcal{U}$ and for any $G_i \in \mathcal{G}$, the multi-granularity fuzzy neighborhood relative cardinality of $[G_i]_{\mathcal{B}}^*$ is defined as

$$RC([G_i]_{\mathcal{B}}^*) = |[G_i]_{\mathcal{B}}^*| - \frac{\sum_{[G_j]_{\mathcal{B}}^* \in M'(\mathcal{R}_{\mathcal{B}}^*)} |[G_j]_{\mathcal{B}}^*|}{|M'(\mathcal{R}_{\mathcal{B}}^*)|}, \quad (15)$$

where $|M'(\mathcal{R}_{\mathcal{B}}^*)|$ denotes the number of information granules in $M'(\mathcal{R}_{\mathcal{B}}^*)$ based on $\mathcal{G} - \{G_i\}$.

In the above definition, we utilize relative cardinality to more effectively distinguish whether a multi-granularity fuzzy neighborhood information granule belongs to the majority or minority class. A higher relative cardinality indicates greater similarity between the current granular-ball and most other granular-balls in $[G_i]_{\mathcal{B}}^*$, suggesting that $[G_i]_{\mathcal{B}}^*$ is more likely to belong to the majority class. Conversely, a lower relative cardinality implies lower similarity with most other granular-balls in $[G_i]_{\mathcal{B}}^*$, indicating that $[G_i]_{\mathcal{B}}^*$ is more likely to belong to the minority class. When $RC([G_i]_{\mathcal{B}}^*) \leq 0$, it indicates that G_i belongs to a minority class, suggesting a higher outlier degree. Conversely, when $RC([G_i]_{\mathcal{B}}^*) > 0$, it indicates that G_i belongs to a majority class, implying a lower probability of being an outlier.

By fusing the multi-granularity fuzzy neighborhood relative entropy and fuzzy neighborhood relative cardinality, we can define the deviation degrees of each multi-granularity fuzzy neighborhood information granule as follows.

Definition 11. Given the granular-balls $\mathcal{G} = \{G_1, G_2, \dots, G_g\}$ generated on $\mathcal{U}$ and for any $G_i \in \mathcal{G}$, the deviation degree of $[G_i]_{\mathcal{B}}^*$ is defined as

$$OD([G_i]_{\mathcal{B}}^*) = \begin{cases} RE_{\mathcal{B}}^*(G_i) \cdot \left(\frac{|\mathcal{G}| - RC([G_i]_{\mathcal{B}}^*)}{2|\mathcal{G}|} \right), & RC([G_i]_{\mathcal{B}}^*) > 0, \\ RE_{\mathcal{B}}^*(G_i) \cdot \sqrt{\frac{|\mathcal{G}| + |RC([G_i]_{\mathcal{B}}^*)|}{2|\mathcal{G}|}}, & RC([G_i]_{\mathcal{B}}^*) \leq 0. \end{cases} \quad (16)$$

In the definition of the deviation degree for $[G_i]_{\mathcal{B}}^*$, if $RC([G_i]_{\mathcal{B}}^*) > 0$, indicating that G_i belongs to a majority class, we calculate the deviation degree of $[G_i]_{\mathcal{B}}^*$ using $\left(\frac{|\mathcal{G}| - RC([G_i]_{\mathcal{B}}^*)}{2|\mathcal{G}|} \right)$, resulting in a relatively smaller value. Conversely, if $RC([G_i]_{\mathcal{B}}^*) \leq 0$, indicating that G_i belongs to a minority class, we calculate the deviation degree of $[G_i]_{\mathcal{B}}^*$ using $\sqrt{\frac{|\mathcal{G}| + \text{abs}(RC([G_i]_{\mathcal{B}}^*))}{2|\mathcal{G}|}}$, resulting in a relatively larger value. The square-root function is adopted to provide a moderate non-linear enhancement for minority or sparse granular-balls, where outliers are more likely to appear. Compared with a direct linear amplification, this scaling increases the sensitivity to minority structures while avoiding overly aggressive enlargement of the deviation degree.

However, the above discussion only addresses the deviation degree of $[G_i]_{\mathcal{B}}^*$. To more comprehensively reflect the outlier degrees of granular-balls and samples, we leverage the concept of information fusion by calculating the outlier scores of each granular-ball through a weighted fusion of the deviation

degrees of multiple multi-granularity fuzzy neighborhood information granules, derived from the multi-granularity fuzzy neighborhood relative entropy across various attribute subsets. To avoid the impractical time complexity of considering all attribute subsets, we only fuse the uncertainty information from each individual attribute in $\mathcal{C}$ to calculate the final outlier scores.

Definition 12. Given the granular-balls $\mathcal{G} = \{G_1, G_2, \dots, G_g\}$ generated on $\mathcal{U}$ and for any $G_i \in \mathcal{G}$, the outlier score of G_i is defined as

$$OS(G_i) = 1 - \frac{\sum_{c \in \mathcal{C}} (1 - OD([G_i]_{\mathcal{C}}^*)) \times W_c(G_i)}{|\mathcal{C}|}, \quad (17)$$

where $W_c(G_i) = \sqrt{\frac{|[G_i]_{\mathcal{C}}^*|}{|\mathcal{G}|}}$ denotes the weight coefficient of G_i w.r.t. $\mathcal{R}_{\mathcal{C}}^*$.

We calculate the weighted fusion of the deviation degrees of the multiple multi-granularity fuzzy neighborhood information granules associated with G_i using $W_c(G_i)$. When $|[G_i]_{\mathcal{C}}^*|$ is relatively small, we consider G_i to belong to a minority class in $\mathcal{G}$ under $\mathcal{R}_{\mathcal{C}}^*$, assigning a smaller weight to G_i , which results in a larger outlier score under $\mathcal{R}_{\mathcal{C}}^*$. The multi-granularity fuzzy neighborhood relative entropy fusion strategy in MFNE effectively mines uncertainty information in data by integrating fuzzy neighborhood relative entropy across different granularities. This strategy fully leverages the representational power of multi-granularity granular-balls and combines the strengths of FNRS to comprehensively capture potential outlier patterns in data. By doing so, it not only reduces the sensitivity to noise inherent in single-granularity paradigms but also enhances adaptability to complex data distributions, thereby improving the effectiveness and robustness of outlier detection.

It should be noted that MFNE does not manually select different fusion rules for different dataset types. The fusion process is data-driven: the adaptive granular-ball generation first determines the granularity structure according to the data distribution, and the fuzzy neighborhood relative entropy, relative cardinality, and weight coefficient $W_c(G_i)$ then jointly determine the contribution of each attribute-induced information granule. Therefore, different datasets affect the final fusion result through their generated granular-ball structures and entropy-based deviation degrees, without requiring a manually designed fusion rule.

After obtaining the granular-ball-level outlier scores, MFNE assigns $OS(G_i)$ to all samples in G_i , i.e., $OS(x) = OS(G_i)$ for any $x \in G_i$. This strategy improves computational efficiency, but may lose sample-level precision when a granular-ball contains both normal samples and small outlier micro-clusters. Although the adaptive granular-ball generation process can partially alleviate this issue by further dividing low-quality or overly large granular-balls, uniform assignment is still an approximation. In future work, ball-internal refinement or sample-level correction can be introduced for boundary granular-balls.

B. Detection algorithm

Based on the proposed outlier detection method, in this section, we introduce the MFNE algorithm, providing its

pseudocode and time complexity analysis.

Algorithm 2: MFNE algorithm

Input: Data matrix U , conditional attribute set $\mathcal{C}$
Output: Outlier score OS

```

1 Generate multi-granularity granular-balls  $\mathcal{G}$  on  $U$  by
  Algorithm 1;
2 for  $c \in \mathcal{C}$  do
3   Construct the multi-granularity fuzzy neighborhood
     granular structure  $M(\mathcal{R}_c^*)$  by Eq. (8);
4   Calculate the multi-granularity fuzzy neighborhood
     entropy  $EN(c)$  by Eq. (10);
5 end
6 for  $G_i \in \mathcal{G}$  do
7   for  $c \in \mathcal{C}$  do
8     Calculate the multi-granularity fuzzy
       neighborhood entropy  $EN_{G_i}(c)$  by Eq. (13);
9     Calculate the multi-granularity fuzzy
       neighborhood relative entropy  $RE_c^*(G_i)$  by
       Eq. (14);
10    Calculate the multi-granularity fuzzy
      neighborhood relative cardinality  $RC([G_i]_c^*)$ 
      by Eq. (15);
11    Calculate the deviation degree  $OD([G_i]_c^*)$  by
      Eq. (16);
12  end
13  Calculate the outlier score  $OS(G_i)$  by Eq. (17);
14  Map  $OS(G_i)$  to samples inside  $G_i$ ;
15  Remove  $OS(G_i)$  from  $OS$ ;
16 end
17 return  $OS$ .
```

The MFNE algorithm is presented in Algorithm 2. It begins by generating multi-granularity granular-balls $\mathcal{G}$ from the input data matrix U , creating a multi-granularity representation of the original dataset. Next, on $\mathcal{G}$, we construct the multi-granularity fuzzy neighborhood granular structure $M(\mathcal{R}_c^*)$ w.r.t. different attributes and calculate the fuzzy neighborhood entropy $EN(c)$. Subsequently, for each granular-ball, we calculate the multi-granularity fuzzy neighborhood entropy, fuzzy neighborhood relative entropy $RE_c^*(G_i)$, fuzzy neighborhood relative cardinality $RC([G_i]_c^*)$, and deviation degree $OD([G_i]_c^*)$. The outlier score for each granular-ball is then calculated by weighted fusion of the deviation degrees of the multiple multi-granularity fuzzy neighborhood information granules to which the granular-ball belongs. Finally, the outlier scores of the granular-balls are mapped back to the samples within each granular-ball to obtain individual sample outlier scores.

The computational complexity of MFNE is further analyzed as follows. Let n denote the number of samples, $|C|$ denote the number of conditional attributes, and $|G|$ denote the number of generated granular-balls. The granular-ball generation process requires $O(n \log n)$ time. After the original samples are represented by granular-balls, MFNE constructs the multi-granularity fuzzy neighborhood granular structure for each attribute. Since the fuzzy relation matrix among granular-

balls needs to be calculated under each attribute, this step requires $O(|C||G|^2)$ time. For each granular-ball and each attribute, MFNE further calculates the multi-granularity fuzzy neighborhood entropy after removing the current granular-ball, the corresponding relative entropy, relative cardinality, and deviation degree. In the worst case, this part also requires $O(|C||G|^2)$ time. The weighted fusion of deviation degrees over all attributes requires $O(|C||G|)$, and mapping the granular-ball-level outlier scores back to sample-level scores requires $O(n)$. Therefore, the overall time complexity of MFNE is $O(n \log n + |C||G|^2 + |C||G| + n)$, which is dominated by $O(n \log n + |C||G|^2)$. In contrast, traditional FNRS-based outlier detection methods usually construct fuzzy relation matrices at the sample level, resulting in a complexity of $O(|C|n^2)$. Since the number of generated granular-balls $|G|$ is generally much smaller than the number of samples n , the proposed MFNE can reduce the computational burden while preserving multi-granularity information for outlier detection.

V. EXPERIMENTS

A. Experimental datasets

We collect 16 datasets commonly used for evaluating outlier detection methods from online sources¹, with their basic information presented in Table I. To mitigate the impact of differing scales across attributes, we apply min-max normalization to preprocess the original datasets. Subsequently, we remove the label information from the datasets to create an unsupervised task environment.

TABLE I: Basic information of 16 experimental datasets.

Datasets	Abbr.	# Samples	# Attributes	# Outliers
Diabetes_tested_positive_26_variant1	Diab	526	8	26
Iris_Irisvirginica_11_variant1	Iris	111	4	11
Mammography	Mam	11183	6	260
Pendigits	Pend	6870	17	156
Pima_TRUE_55_variant1	Pima	555	9	55
Satellite	Sate	6435	37	2036
Sonar_M_10_variant1	Sonar	107	60	10
Spambase_spam_56_variant1	Spam	2844	58	56
Thyroid	Thyroid	3772	6	93
Wbc_malignant_39_variant1	Wbc	483	9	39
Wdbc_M_39_variant1	Wdbc	396	31	39
Yeast_ERL_5_variant1	Yeast	1141	8	5
CreditA_plus_42_variant1	Credit	425	15	42
German_1_14_variant1	German	714	20	14
Heart270_2_16_variant1	Heart	166	13	16
Horse_1_12_variant1	Horse	256	27	12

B. Experimental settings

We select 14 unsupervised outlier detection methods, encompassing recently proposed GBC-based methods and effective classical methods. These methods include LOF [42], Outlier Detection using Indegree Number (ODIN) [43], Local Distance-based Outlier Factor (LDof) [44], Local Outlier Probabilities (LoOP) [45], reverse uNreaChability-based outlier detection (NC) [46], provable Self-Representation based Outlier detection (SRO) [47], Deep Support Vector Data Description (DeepSVDD) [11], joint learning-based outlier

¹<https://github.com/BELLoney/Outlier-detection>

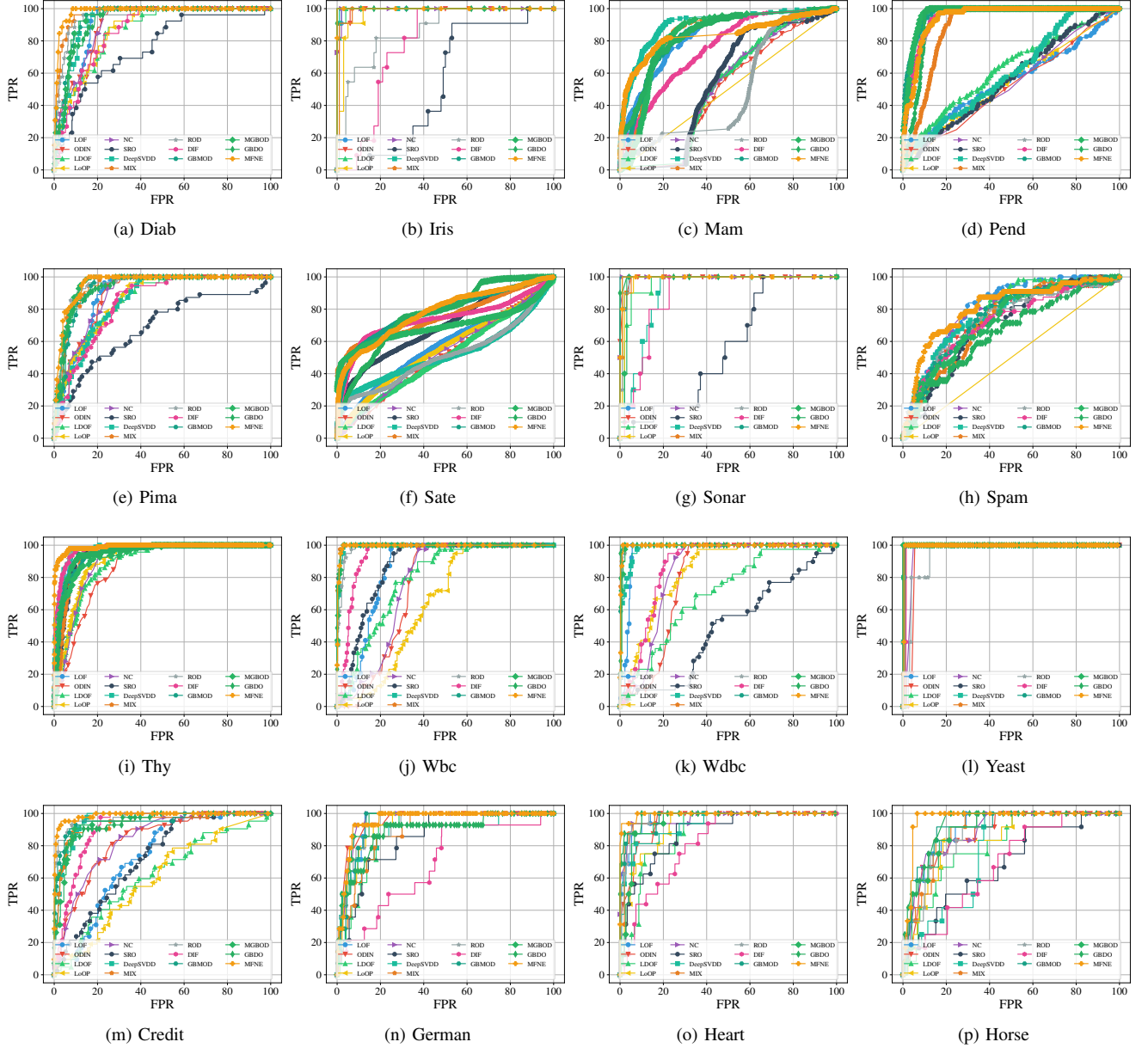


Fig. 2: ROC curves of 14 methods across 16 datasets for outlier detection.

detector (MIX) [48], Rotation-based Outlier Detection (ROD) [49], Deep Isolation Forest for outlier detection (DIF) [4], Granular-Ball Mean-shift Outlier Detector (GBMOD) [21], Multi-scale Granular-Balls-based Outlier Detection (MGBOD) [5], and local Granular-Ball Density-based Outlier detection (GBDO) [10].

For methods requiring manual hyperparameter tuning, we set reasonable search ranges and select the optimal results for each dataset as the final experimental outcomes. For LOF, ODIN, LDOF, LoOP, and NC, the number of neighbors for each object is searched from 2 to 60 with a step size of 1. For SRO, the hyperparameter α controlling the optimization term is searched from 2 to 20 with a step size of 1. For GBMOD and GBDO, the number of granular-ball neighbors for each object is searched from 2 to $\min(60, |\mathcal{G}|)$ with a step size of 1, where

$|\mathcal{G}|$ is the number of generated granular-balls. For MGBOD, the fuzzy radius is searched from 0.1 to 2 with a step size of 0.1. For GBRAD, the granular-ball splitting parameter is searched from 3 to 40 with a step size of 1. For our proposed method MFNE, the hyperparameter σ is searched from 0.1 to 2 with a step size of 0.1. The sensitivity analysis of MFNE w.r.t. the hyperparameter σ will be presented in the subsequent experimental results.

To provide a comprehensive evaluation, we employ the Receiver Operating Characteristic (ROC) curve, Area Under the Curve (AUC), Average Precision (AP), and Geometric mean (G-mean) [10], [20], [50]–[52]. AUC and AP are calculated directly from the continuous outlier scores and do not require a manually specified fixed threshold. AUC is obtained from the ROC curve over all candidate thresholds induced by the outlier

TABLE II: AUC scores of 14 methods across 16 datasets, with the best results highlighted in **bold**.

Datasets	LOF	ODIN	LDOF	LoOP	NC	SRO	DeepSVDD	MIX	ROD	DIF	GBMOD	MGBOD	GBDO	MFNE (ours)
Diab	0.905	0.898	0.863	0.870	0.902	0.754	0.929	0.969	0.954	0.861	0.945	0.939	0.926	0.983
Iris	0.998	0.993	0.995	0.969	0.996	0.534	0.999	0.986	0.866	0.779	1.000	1.000	1.000	1.000
Mam	0.833	0.490	0.523	0.500	0.514	0.544	0.905	0.832	0.497	0.755	0.830	0.834	0.840	0.836
Pend	0.551	0.517	0.628	0.550	0.537	0.562	0.603	0.886	0.948	0.952	0.968	0.970	0.935	0.944
Pima	0.891	0.889	0.855	0.860	0.888	0.685	0.849	0.945	0.947	0.839	0.935	0.954	0.934	0.960
Sate	0.568	0.515	0.510	0.559	0.524	0.694	0.522	0.756	0.513	0.735	0.781	0.696	0.768	0.780
Sonar	0.994	0.994	0.964	0.981	0.998	0.562	0.901	0.989	0.990	0.882	1.000	0.975	0.998	0.995
Spam	0.803	0.746	0.762	0.500	0.767	0.702	0.756	0.730	0.736	0.718	0.748	0.725	0.661	0.809
Thy	0.955	0.863	0.887	0.909	0.903	0.946	0.976	0.941	0.977	0.975	0.953	0.937	0.951	0.990
Wbc	0.849	0.734	0.792	0.645	0.756	0.870	0.993	0.997	0.985	0.939	0.997	0.994	0.997	0.996
Wdbc	0.964	0.786	0.702	0.839	0.834	0.489	0.981	0.996	0.995	0.871	0.996	0.991	0.997	0.997
Yeast	0.992	0.954	0.987	0.988	0.970	0.995	0.997	1.000	0.949	0.987	1.000	0.999	1.000	0.996
Credit	0.726	0.832	0.616	0.598	0.837	0.724	0.959	0.937	0.974	0.910	0.955	0.941	0.948	0.989
German	0.941	0.952	0.889	0.925	0.955	0.852	0.882	0.891	0.927	0.686	0.937	0.950	0.889	0.955
Heart	0.973	0.949	0.880	0.923	0.969	0.880	0.930	0.974	0.975	0.817	0.973	0.947	0.985	0.995
Horse	0.902	0.870	0.803	0.838	0.885	0.685	0.761	0.893	0.880	0.683	0.909	0.893	0.928	0.967
Average	0.865	0.811	0.791	0.778	0.827	0.717	0.872	0.920	0.882	0.837	0.933	0.922	0.922	0.950

TABLE III: AP scores of 14 methods across 16 datasets, with the best results highlighted in **bold**.

Datasets	LOF	ODIN	LDOF	LoOP	NC	SRO	DeepSVDD	MIX	ROD	DIF	GBMOD	MGBOD	GBDO	MFNE (ours)
Diab	0.297	0.268	0.247	0.266	0.271	0.151	0.321	0.578	0.481	0.216	0.437	0.362	0.378	0.727
Iris	0.986	0.951	0.968	0.742	0.971	0.107	0.992	0.857	0.429	0.200	1.000	1.000	1.000	1.000
Mam	0.118	0.021	0.024	0.023	0.022	0.024	0.293	0.075	0.062	0.112	0.089	0.068	0.148	0.335
Pend	0.051	0.024	0.059	0.045	0.026	0.039	0.044	0.101	0.337	0.319	0.422	0.466	0.151	0.304
Pima	0.401	0.395	0.359	0.373	0.395	0.235	0.347	0.594	0.591	0.304	0.537	0.616	0.544	0.670
Sate	0.397	0.329	0.350	0.385	0.337	0.588	0.481	0.734	0.421	0.676	0.705	0.705	0.622	0.701
Sonar	0.953	0.934	0.711	0.804	0.981	0.115	0.425	0.904	0.887	0.329	1.000	0.701	0.977	0.946
Spam	0.053	0.049	0.044	0.020	0.053	0.038	0.074	0.045	0.091	0.070	0.066	0.048	0.046	0.091
Thy	0.303	0.099	0.161	0.178	0.122	0.205	0.556	0.227	0.485	0.524	0.324	0.312	0.305	0.831
Wbc	0.212	0.133	0.186	0.104	0.144	0.251	0.925	0.967	0.825	0.412	0.965	0.931	0.961	0.942
Wdbc	0.590	0.200	0.177	0.294	0.241	0.110	0.805	0.918	0.894	0.298	0.915	0.890	0.942	0.960
Yeast	0.244	0.064	0.165	0.171	0.069	0.315	0.464	1.000	0.055	0.159	1.000	0.696	1.000	0.417
Credit	0.218	0.348	0.141	0.157	0.339	0.195	0.696	0.702	0.866	0.444	0.829	0.636	0.776	0.937
German	0.186	0.230	0.100	0.155	0.212	0.118	0.210	0.315	0.287	0.062	0.214	0.238	0.292	0.323
Heart	0.833	0.689	0.379	0.590	0.849	0.486	0.675	0.796	0.811	0.269	0.878	0.574	0.912	0.965
Horse	0.295	0.207	0.149	0.172	0.232	0.110	0.154	0.301	0.371	0.118	0.303	0.233	0.410	0.515
Average	0.384	0.309	0.264	0.280	0.329	0.193	0.466	0.570	0.493	0.282	0.605	0.530	0.591	0.666

TABLE IV: G-mean scores of 14 methods across 16 datasets, with the best results highlighted in **bold**.

Datasets	LOF	ODIN	LDOF	LoOP	NC	SRO	DeepSVDD	MIX	ROD	DIF	GBMOD	MGBOD	GBDO	MFNE (ours)
Diab	0.897	0.865	0.815	0.818	0.875	0.694	0.908	0.934	0.914	0.807	0.905	0.906	0.883	0.959
Iris	0.990	0.970	0.980	0.938	0.990	0.654	0.995	0.964	0.819	0.794	1.000	1.000	1.000	1.000
Mam	0.759	0.543	0.561	0.000	0.559	0.609	0.850	0.806	0.522	0.680	0.776	0.803	0.780	0.804
Pend	0.551	0.515	0.600	0.546	0.503	0.545	0.553	0.870	0.904	0.907	0.922	0.942	0.943	0.904
Pima	0.860	0.848	0.786	0.794	0.843	0.645	0.778	0.892	0.902	0.799	0.902	0.921	0.875	0.922
Sate	0.563	0.517	0.506	0.548	0.528	0.632	0.543	0.722	0.526	0.737	0.724	0.710	0.722	0.707
Sonar	0.979	0.974	0.925	0.969	0.995	0.586	0.902	0.974	0.979	0.879	1.000	0.984	0.984	0.995
Spam	0.766	0.702	0.715	0.000	0.743	0.672	0.711	0.694	0.699	0.674	0.708	0.676	0.626	0.755
Thy	0.908	0.813	0.823	0.853	0.867	0.917	0.938	0.902	0.940	0.931	0.894	0.871	0.912	0.955
Wbc	0.861	0.781	0.751	0.663	0.777	0.844	0.982	0.986	0.957	0.926	0.985	0.984	0.983	0.986
Wdbc	0.972	0.827	0.671	0.792	0.837	0.548	0.953	0.993	0.986	0.860	0.992	0.986	0.999	0.993
Yeast	0.996	0.972	0.993	0.994	0.975	0.996	0.997	1.000	0.936	0.992	1.000	0.999	1.000	0.998
Credit	0.687	0.773	0.604	0.603	0.770	0.668	0.921	0.899	0.934	0.876	0.917	0.905	0.884	0.960
German	0.934	0.898	0.866	0.879	0.928	0.789	0.841	0.838	0.866	0.693	0.930	0.924	0.852	0.929
Heart	0.927	0.909	0.841	0.876	0.939	0.833	0.868	0.939	0.942	0.754	0.920	0.904	0.949	0.965
Horse	0.842	0.810	0.758	0.824	0.819	0.641	0.789	0.845	0.815	0.651	0.880	0.849	0.894	0.967
Average	0.843	0.795	0.762	0.694	0.809	0.704	0.845	0.891	0.853	0.810	0.904	0.898	0.893	0.925

scores, while AP is calculated from the ranked scores based on the precision-recall curve. For the threshold-dependent G-mean metric, we compute the G-mean under each candidate threshold on the ROC curve and report the maximum value, i.e., $G\text{-mean} = \max_{\xi} \sqrt{\text{TPR}(\xi) \times \text{TN}(\xi)}$, where $\text{TPR}(\xi)$ and $\text{TN}(\xi)$ denote the true positive rate and true negative rate under the threshold ξ , respectively. The labels are used only for metric calculation, not for model training or outlier

score generation.

C. Experimental results and analysis

In this section, we evaluate the outlier detection performance of 14 methods across 16 datasets using ROC curves, AUC scores, AP scores, and G-mean scores, followed by an analysis of the experimental results. Additionally, we conduct

statistical tests on the results and analyze the sensitivity of MFNE to its hyperparameter σ .

1) *Effectiveness analysis*: As illustrated in Fig. 2, we present the ROC curves of 14 methods across 16 datasets. The figure shows that MFNE's curves are closest to the top-left corner of the axes on datasets such as Diab, Pima, Spam, Thy, Wdbc, Credit, and German, indicating superior performance on these datasets. Since ROC curves make it challenging to discern precise performance differences, we further provide a tabular comparison of AUC scores, AP scores, and G-mean scores for all methods across the 16 datasets. As shown in Table II, MFNE achieves the highest AUC score on 10 datasets and the best average AUC score. Table III displays the AP scores, where MFNE secures the top AP score on 10 datasets and the highest average AP score, surpassing the second-best method by 6.1%. Finally, Table IV presents the G-mean scores, with MFNE achieving the highest G-mean score on 8 datasets and the best average G-mean score, exceeding the second-best method by 2.1%. These results consistently demonstrate the effectiveness of MFNE.

Overall, our proposed MFNE method integrates GBC into FNRS, replacing single-granularity samples with multi-granularity granular-balls as the basic processing unit. By introducing a novel GrC model and new uncertainty measures, along with an innovative multi-granularity fuzzy neighborhood relative entropy fusion strategy, MFNE effectively leverages multi-granularity and uncertainty information in data, substantially enhancing its outlier detection performance across datasets of varying scales.

2) *Runtime analysis*: To further evaluate the practical feasibility of MFNE, we report the running time of different outlier detection methods on the 16 datasets, as shown in Table V. The running time is measured in seconds under the same experimental environment, excluding data loading and preprocessing. For granular-ball-based methods, the reported running time includes the granular-ball generation process and the subsequent outlier score calculation.

From Table V, MFNE achieves an average running time of 0.955 seconds. Although MFNE is not the fastest method among all compared methods, its computational cost remains within a practical range on all datasets. Specifically, MFNE is faster than several representative methods, including SRO, DeepSVDD, MIX, ROD, DIF, GBMOD, and MGBOD in terms of average running time. Compared with GBC-based methods, MFNE is slower than GBDO on average, mainly because MFNE calculates fuzzy neighborhood entropy-related measures under multiple attributes, but it is faster than GBMOD and MGBOD. These results show that MFNE provides a reasonable trade-off between detection performance and computational efficiency.

3) *Statistical analysis*: To assess whether the performance differences among different outlier detection methods are statistically significant, we conduct the Friedman test followed by the Nemenyi post-hoc test on the AUC scores across all datasets [10], [20], [50]. The Friedman test ranks the methods on each dataset and compares their average ranks, while the Nemenyi test further determines whether the rank differences between method pairs exceed the Critical Difference (CD)

under the selected significance level. The results are presented using a CD diagram, where each method is placed according to its average rank. Methods connected by the same horizontal line are not significantly different.

In the experiments of this study, involving 14 methods and 16 datasets, the degrees of freedom for the τ_F distribution are calculated as 13 and 195. Using a significance level of $\alpha = 0.1$ in the Friedman test, we obtain a τ_F value of 13.7194, which exceeds the critical value of 1.5585, indicating statistically significant differences among the methods, necessitating the Nemenyi test for further pairwise differentiation. At $\alpha = 0.1$, we calculate the Critical Difference (CD) value as $CD_{0.1} = 4.6145$ and present the Nemenyi test results in Fig. 3. The figure shows that MFNE's line segment overlaps only with those of GBMOD, GBDO, MGBOD, MIX, and LOF, while exhibiting statistically significant differences with the remaining 8 methods, effectively demonstrating MFNE's superior outlier detection performance.

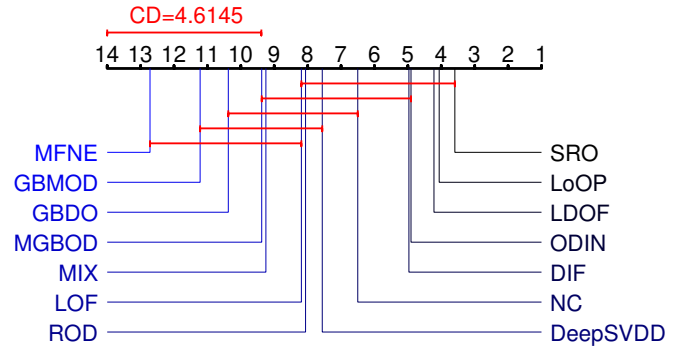


Fig. 3: Nemenyi test diagram comparing statistical differences of 14 methods across 16 datasets.

4) *Hyperparameter sensitivity analysis*: As outlined in Definition 5, the hyperparameter σ controls the size of the fuzzy neighborhood radius through $\varepsilon_B = \text{std}(\mathcal{B})/\sigma$. To examine its influence on MFNE, we analyze the AUC variation of MFNE on 16 datasets by varying σ from 0.1 to 2.0 with a step size of 0.1.

As shown in Fig. 4, the performance of MFNE varies with different values of σ , but remains relatively stable within a broad range after σ reaches a moderate value. When σ is small, the corresponding neighborhood radius is relatively large, which may produce coarse fuzzy neighborhood information granules and weaken the discrimination of local abnormal structures. As σ increases, the neighborhood radius becomes smaller, allowing MFNE to construct more localized fuzzy neighborhood granules and capture finer distributional differences. However, an excessively large σ may also make the neighborhood too narrow and reduce the amount of available neighborhood information. Therefore, a moderate value of σ is preferred to balance locality and neighborhood information.

Overall, the sensitivity results indicate that MFNE is not overly sensitive to σ within the tested range. In our experiments, $\sigma = 0.4$ provides a favorable overall trade-off across the evaluated datasets and is used as a representative setting in the sensitivity analysis.

TABLE V: Runtime comparison of different outlier detection methods on 16 datasets. The running time is reported in seconds, and the fastest method on each dataset as well as the lowest average running time are highlighted in **bold**.

Datasets	LOF	ODIN	LDOF	LoOP	NC	SRO	DeepSVDD	MIX	ROD	DIF	GBMOD	MGBOD	GBDO	MFNE (ours)
Diab	0.006	0.015	0.126	0.005	0.216	1.097	3.756	2.314	4.077	5.275	0.110	0.090	0.539	2.279
Iris	0.001	0.002	0.021	0.001	0.003	0.113	3.047	0.394	0.014	0.504	0.667	0.012	0.041	0.022
Mam	0.153	3.506	4.591	0.183	0.712	218.838	11.671	5.039	3.316	11.726	3.755	8.610	0.528	5.379
Pend	0.870	1.778	3.638	0.491	0.403	123.121	8.680	17.457	60.337	9.220	2.704	8.484	0.472	2.191
Pima	0.004	0.016	0.112	0.013	0.017	1.199	3.369	1.178	0.655	0.927	0.141	0.100	0.097	0.081
Sate	0.043	1.158	3.132	0.071	0.542	121.104	8.111	17.919	401.724	8.483	3.322	11.345	0.519	2.507
Sonar	0.008	0.007	0.022	0.005	0.003	0.076	2.751	0.867	74.484	0.490	0.216	0.043	0.035	0.012
Spam	0.052	0.310	0.778	0.065	0.212	25.714	4.942	1.422	794.536	4.147	7.618	6.084	0.625	1.445
Thy	0.046	0.434	1.245	0.069	0.157	40.146	5.670	3.269	1.123	4.735	1.034	1.639	0.281	0.841
Wbc	0.004	0.012	0.045	0.010	0.013	0.940	3.097	0.567	0.218	0.816	0.115	0.050	0.134	0.052
Wdbc	0.008	0.027	0.078	0.007	0.013	0.750	3.004	18.956	30.259	0.712	0.797	0.101	0.079	0.058
Yeast	0.015	0.089	0.246	0.020	0.037	4.787	3.747	1.511	0.628	1.871	0.276	0.267	0.140	0.172
Credit	0.007	0.010	0.081	0.006	0.014	0.656	2.915	8.010	2.280	0.726	0.101	0.098	0.081	0.071
German	0.010	0.045	0.145	0.012	0.025	1.444	3.597	4.331	4.352	1.369	1.440	0.283	0.109	0.110
Heart	0.002	0.003	0.030	0.002	0.005	0.130	3.033	2.756	0.608	0.545	0.321	0.033	0.046	0.028
Horse	0.004	0.009	0.047	0.006	0.007	0.261	3.272	7.462	7.028	0.621	0.506	0.064	0.077	0.040
Average	0.077	0.464	0.896	0.060	0.149	33.774	4.666	5.841	86.602	3.261	1.445	2.331	0.238	0.955

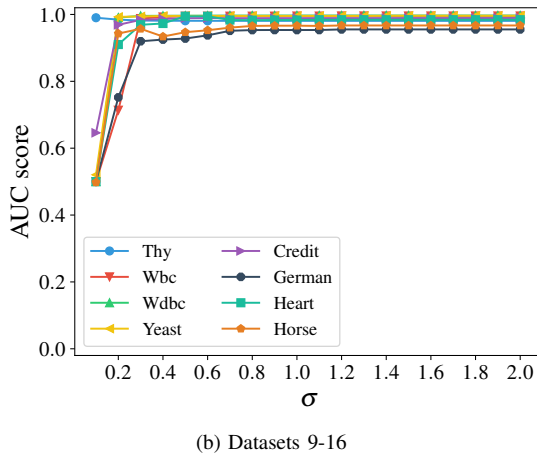
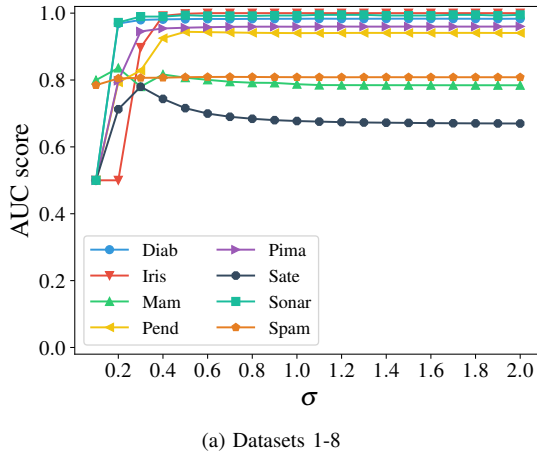


Fig. 4: Sensitivity analysis of MFNE's AUC scores across 16 datasets with varying σ .

VI. CONCLUSIONS

In this study, we propose a novel unsupervised outlier detection method, termed MFNE, which integrates GBC and FNRS. MFNE adopts multi-granularity granular balls as fundamental processing units to construct a multi-granularity fuzzy neighborhood rough set model and introduces several new

uncertainty measures. Furthermore, a novel multi-granularity information fusion strategy is designed to effectively exploit both multi-granularity structures and uncertainty information inherent in the data. Extensive experiments demonstrate that MFNE achieves superior performance across multiple evaluation metrics, consistently outperforming 13 baseline methods. Despite its strong performance, MFNE relies on hyperparameter tuning, which may affect its robustness across different datasets and pose challenges for non-expert users. Future work will focus on developing adaptive detection mechanisms and incorporating richer uncertainty measures to further enhance detection performance.

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